



Worm Brook 2023-4

A report on behalf of the Wye Salmon Association.

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A year in the life of the Worm Brook, as evidenced by citizen science data recorded between Nov-23 and Oct-24.

The Worm Brook drains 73 km² of Herefordshire farmland, located immediately to the south-west of Hereford. The catchment, shown in Figure 1, is for the most part moderately flat and is dominated by arable agriculture (54%). Improved grassland (31%) and woodland (11%) tend to occupy the steeper slopes along the south-east watershed and around the Jury Brook valley. To the west of the Worm sub catchment lies the steeper sided Dore valley and to the south is located the topologically similar Garren/Gamber sub catchment. There are several very large IPUs within the Worm sub catchment, in addition to more in the upper Garren/Gamber sub catchment to the south. There will undoubtedly be a lot of manure and digestate spreading going on in this area, with consequences for diffuse RP loads.

Sampling locations

WSA sampling locations, as shown in figure 1, are summarised in Table 1.

Reference	Site name	Description
<i>Allensmore Brook:</i>		
WSA030	Worm, The Tram	Sole site
<i>Upper Worm (above the Allensmore confluence):</i>		
WSA129	Worm, below Wormlow STW	Above Allensmore conf., below Much Dewchurch STW
<i>Lower Worm (from the Allensmore confluence down to the Monnow confluence):</i>		
WSA031	Worm, Gough's Bridge	Above Kilpeck STW
WSA142	Worm, below Kilpeck STW	Below Kilpeck STW
WSA143	Worm, Wormbridge	Below Wormbridge STW
WSA032	Worm, Sawmills Footbridge	Most downstream location
<i>Jury Brook:</i>		
WSA156	Worm, Jury Brook	Sole site (not sampled before March-24)

Table 1: summary of WSA sampling sites in the Worm sub catchment.

Point sources

Three Dwr Cymru sewage treatment works are also shown in figure 1 and summarised below in table 2. As noted in our previous report, the situation at Much Dewchurch is worrying. Table 2 shows that there is no spare capacity at this facility and dry weather flow (DWF) is already 'at consent'. Indeed, we previously estimated that dry weather effluent constitutes around a fifth of the Worm's flow immediately downstream of this facility. Moreover, since there is no headroom, we might expect CSO emission to be a problem. They are. In 2023, this facility discharged untreated sewage 64 times, for a total duration of 1,016 hours. That equated to six weeks of continuous discharge.

Site name	Capacity (PE)	Input load (PE)
Much Dewchurch (new) STW	499	499
Kilpeck STW	n/a	80
Wormbridge STW	n/a	50

Table 2: summary of DC-WW sewage treatment plants (2024 FoI req.).

The other two STWs are an order of magnitude smaller and are much further down the sub catchment. Hence, they discharge into much larger flow volumes. But we shall see what our data tells us on their impact.

Flow monitoring

It will be helpful for our analysis later to have volumetric flow data available. Unfortunately, the Environment Agency (AE) do not have flow monitoring on the Worm Brook. There are height stations on the Dore at Peterchurch and on the Dulas at Ewyas Harold, and there is a height/flow station downstream on the Monnow at Grossmont. However, none of these are ideal for our purpose. Only Grossmont is flow-calibrated, and this includes water from the upland, western Monnow catchment, which is unlikely to be representative of the Worm. Instead, we shall use data from the Garren at Marstow Mill, on the basis that an immediately adjacent, lowland catchment is likely to be subject to very similar rainfall patterns to the Worm.

Results and discussion

Summary statistics

The data comprises 286 sets of readings across the seven sites. Six sites have approximately weekly data over the period November-23 to October-24 (typically ~47 each). The seventh site, Sawmills Bridge, which commenced sampling in March-24, has 18 sets of readings for EC and RP.

Mean values for the four parameters are shown in table 3.

Reference	Site name	Turbidity (NTU)	EC (uS/cm)	pH	RP (mgP/l)
WSA030	Worm, The Tram	17	686	7.9	0.11
WSA129	Worm, below Wormlow STW	16	586	7.8	0.26
WSA031	Worm, Gough's Bridge	45	608	7.7	0.14
WSA142	Worm, below Kilpeck STW	24	614	7.7	0.17
WSA143	Worm, Wormbridge	22	600	7.9	0.14
WSA032	Worm, Sawmills Footbridge		515		0.11
WSA156	Worm, Jury Brook	9	631	7.8	0.05

Table 3: annual mean values for turbidity, EC, pH and RP over the period Nov-23 to Oct-24

The above values are consistent with EA data for this river, other than turbidity which EA do not measure. We shall now examine each parameter in more detail.

Electrical conductivity

Electrical conductivity is dominated by dissolved mineral ions. Because of the marl bedrock found across most of the Herefordshire Wye catchment, a major component of the ionic load here is CaCO_3 . This substance dissolves relatively readily, causing electrical conductivity to be relatively high, compared say to the upper, Welsh catchment. Dissolved nutrients (nitrate, phosphate, etc.) typically only make up around 1% of the ionic load, hence EC is not a useful indicator of this type of pollution. Nevertheless, elevated readings can be associated with point sources (STWs, farm drains, etc.) and transient spiles can be indicative of episodic pollution. More generally, EC tends to have a relationship with river flowrate. In wet weather, the water drains to the river more rapidly, often via surface pathways, and has less opportunity to dissolve minerals. Hence EC falls with flowrate.

Figure 2 shows the EC trend plot for the Worm at the Tram Inn. This exhibits generally higher readings between June and September than earlier in the year. There are also four apparently anomalous readings, circled in grey. Figure 3 shows the utility of plotting this data against estimated river flowrate (using the Garren at Marstow Mill as a surrogate). The strong flow-dependency is immediately apparent, with the regression line accounting for approximately half of the data variance. Moreover, three of the suspected outliers now fall within the typical data spread.

The remaining outlier (1322 $\mu\text{S}/\text{cm}$ on 2-August) stands out even more in figure 3. Our expectation for the reading on that day would have been around 800 $\mu\text{S}/\text{cm}$, so there must have been a serious spike of pollution present on that day. Clearly something got into the river that shouldn't have. In addition to general farming activities upstream of this point on the Allensmore Brook, we also find the Allensmore Nurseries and the Cargill feed mill, both within a few hundred metres.

Rather than present seven sets of charts, table 4 below list the dates and magnitudes of observed spikes. We see that a spike was also observed on 2-August below Kilpeck STW, about 5km downstream of the Tram. Another correlated spike occurred on 03-July at Goughs Bridge and Kilpeck STW, these locations being close together. Two sets of correlation like this strongly suggests that these are genuine results rather than typos or mis-readings.

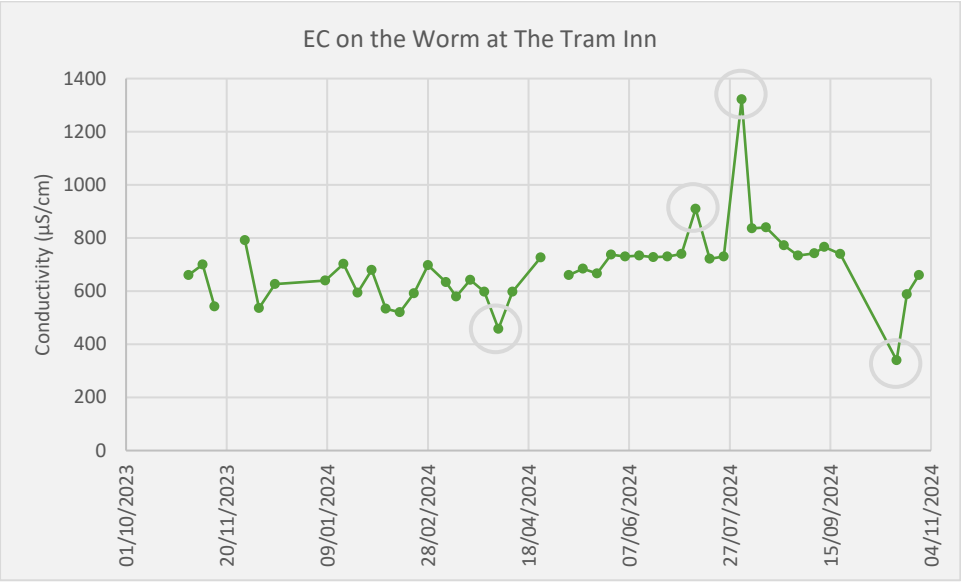


Figure 2: electrical conductivity trend on the Worm Brook at the Tram Inn.

Suspected outliers are circled grey.

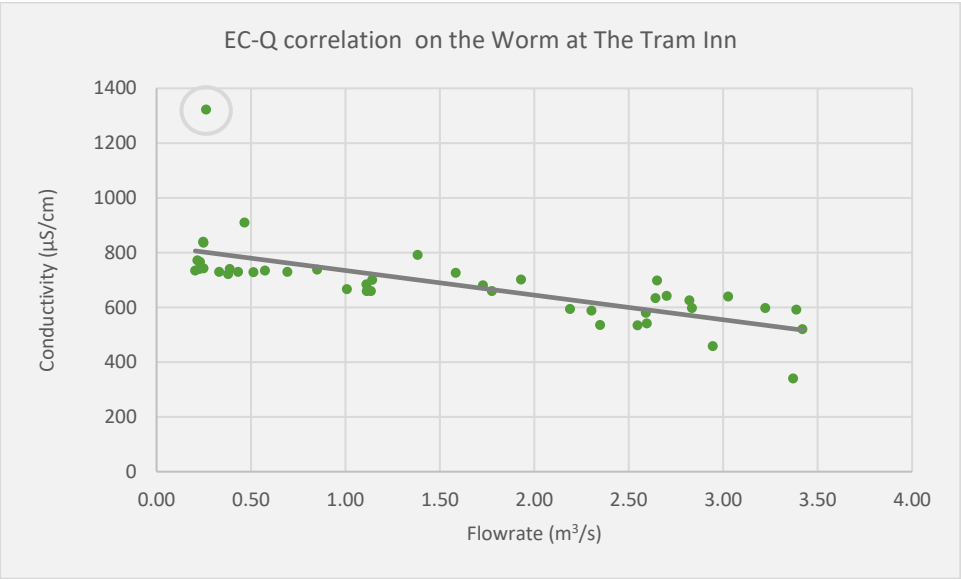


Figure 3: data of figure 2 plotted against flowrate at Marstow Mill in the neighbouring Garren catchment. The grey line is a linear regression to the data ($R^2 = 0.49$).

Only 1 outlier remains.

Reference	Site name	Date	EC (uS/cm)
WSA030	Worm, The Tram	02-Aug	1322
WSA129	Worm, below Wormlow STW	-	-
WSA031	Worm, Gough's Bridge	03-July	780
WSA142	Worm, below Kilpeck STW	03-July	978
		02-Aug	856
WSA143	Worm, Wormbridge	23-Oct	940
WSA032	Worm, Sawmills Footbridge	-	-
WSA156	Worm, Jury Brook	20-Mar	852
		08-May	876

Table 4: summary of EC spikes observed throughout the Worm catchment over the period Nov-23 to Oct-24

pH

pH on the Worm Brook is largely determined by the CaCO_3 load (often referred to as ‘alkalinity’). Consequently, this parameter also has a flowrate dependency, albeit rather weaker than that for EC discussed above. The utility of this parameter again lies in picking up outliers from the natural range. Agricultural causes may include slurry spills (low outliers) and digestate spills (high outliers).

As an exemplar, we shall again look at the Tram Inn sampling point. Figure 4 shows the time trend and figure 5 the flowrate correlation. Three outliers are apparent on the time trend. Because the flowrate

dependency is relatively weak for pH, the R^2 value for the regression line is low. Hence the flowrate correlation is of less utility for pH than for EC.

Table 5 lists outliers found at each sampling site. Only one inter-site correlation is found at the Tram and at Wormbridge on 01-November. However, note that the low pH recorded on the Jury brook on 08-May coincides with an EC spike on the same day (table 4). This is strongly suggestive of episodic point pollution.

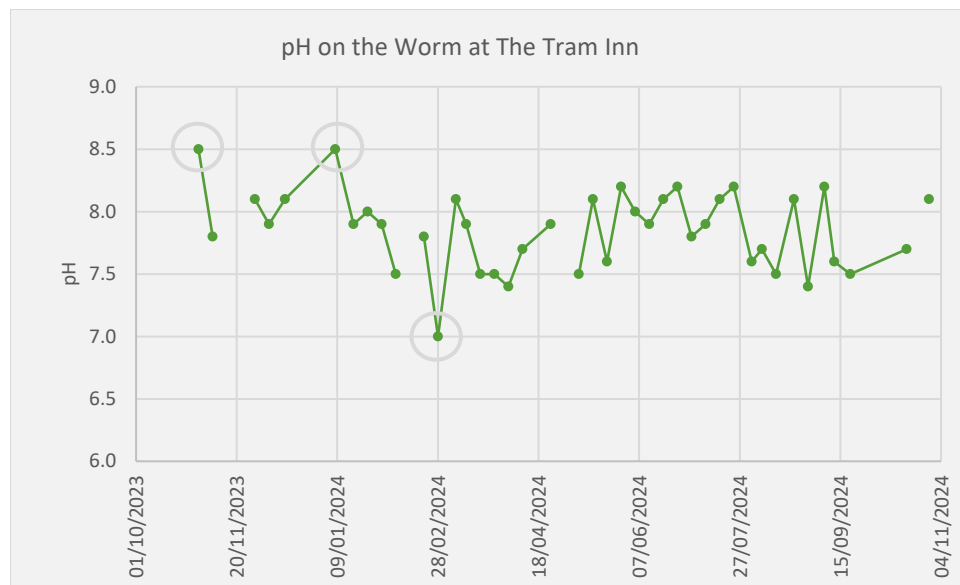


Figure 4: pH trend on the Worm Brook at the Tram Inn.

Suspected outliers are circled grey.

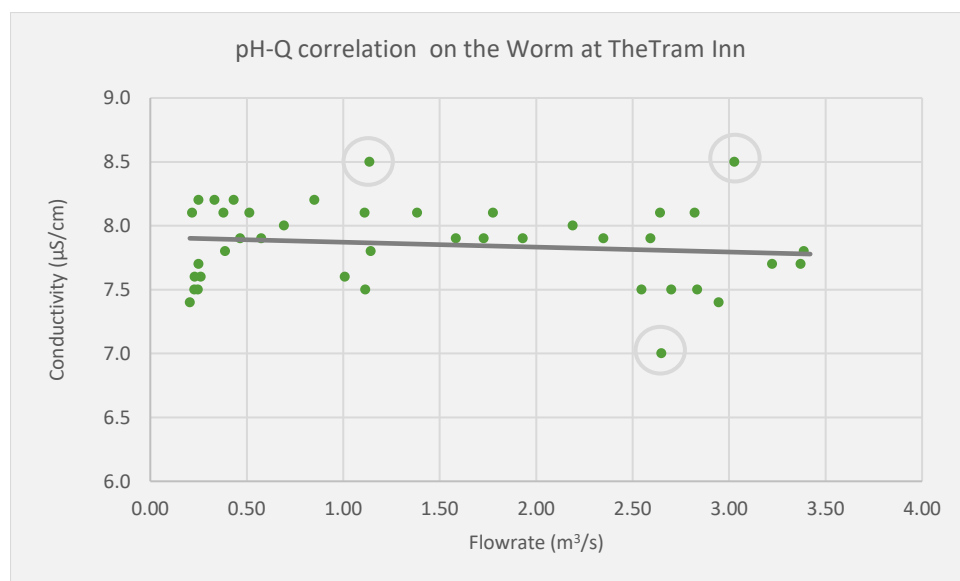


Figure 5: data of figure 4 plotted against flowrate at Marstow Mill in the neighbouring Garren catchment. The grey line is a linear regression to the data ($R^2 = 0.02$).

Suspected outliers are circled grey.

Reference	Site name	Date	pH
WSA030	Worm, The Tram	01-Nov	8.5
		08-Jan	8.5
		28-Feb	7.0
WSA129	Worm, below Wormlow STW	17-Jan	8.5
WSA031	Worm, Gough's Bridge	-	-
WSA142	Worm, below Kilpeck STW	-	-
WSA143	Worm, Wormbridge	01-Nov	8.9
		31-Jan	8.9
WSA032	Worm, Sawmills Footbridge	-	-
WSA156	Worm, Jury Brook	23-Aug	8.5
		08-May	7.1

Table 5: summary of pH outliers observed throughout the Worm catchment over the period Nov-23 to Oct-24.

High values red; low values are blue.

Turbidity

Nephelometric turbidity has a very strong flowrate dependence, so we shall view this data solely as a flowrate correlation. Turbidity measurements have been collected since May-24. So far, this has provided us with a useful low-flow dataset, but a distinct lack of data for high flow conditions. The data presented for Kilpeck in figure 6 is typical and illustrates the point. For this reason, a fuller analysis of turbidity will be deferred until we have accumulated a full year's data.

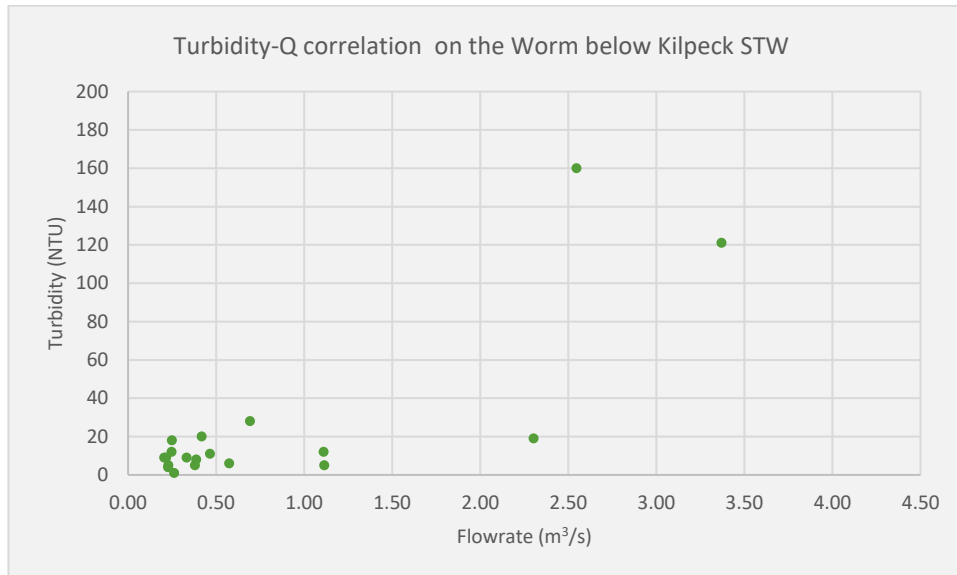


Figure 6: turbidity-flowrate correlation for the Worm below Kilpeck STW.

Nevertheless, it is worth highlighting one measurement recorded at Goughs Bridge on 24-July, when turbidity was recorded as 660 NTU. This occurred in dry weather, with the river at summer levels; the Garren gauging station recorded 0.33 m³/s that day. This appeared to be a localised soil run-off issue, not being recorded at other locations that day and not correlating with spikes in EC or pH. The issue was reported to the EA pollution hotline.

Reactive phosphorus

Reactive phosphorus also has a strong flowrate dependence. This property can be utilised to pull out the relative contributions from fixed-flow sources such as STWs and diffuse, mainly agricultural, sources³.

We begin at the top of the Worm Brook, below Wormlow STW (known to Dwr Cymru as Much Dewchurch STW). The time trend plot (figure 7) is intriguing, with rising RP concentrations from April to September. The reason for this becomes apparent when we examine the flowrate correlation in figure 8. This is simply a dilution curve, with concentration varying as the reciprocal of volumetric flowrate – as indicated by the LAM curve in grey. The STW outputs the same volume of effluent every day, so the resultant concentration of RP in the Worm is inversely proportional to its flowrate. At high flowrates, the signal bottoms out around 0.1 mgP/l, which represents the background diffuse agricultural load.

This fixed-flow RP load from Much Dewchurch STW persists all the way down to Sawmills Footbridge, diluting progressively along the way, but also being topped up measurably by Kilpeck STW.

An estimate of the annual average flowrate at the monitoring point below Wormlow STW of 0.1 m³/s allows us to calculate a fixed-flow RP load of around 1.2kgP/day. This is broadly in line with the expected output from over 500 people.

³ This technique is referred to as load apportionment modelling (LAM).



Figure 7: RP on the Worm below Wormlow STW.

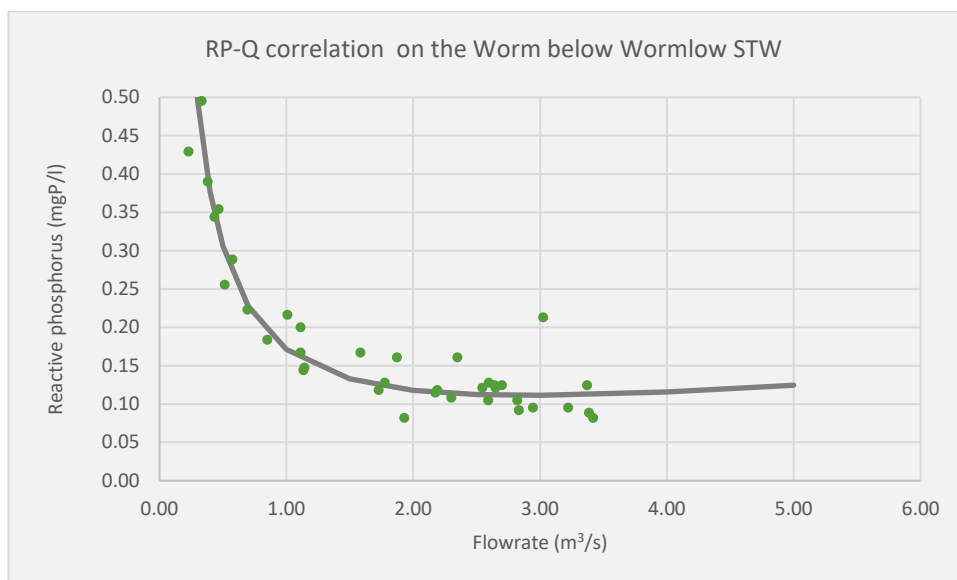


Figure 8: RP-flowrate correlation for the above data. The grey line represents the best-fit LAM model.

In contrast to the above, it is worthwhile examining the plot for the Jury Brook, a very different sub catchment containing no STWs. We find just a slowly rising curve typical of diffuse agricultural pollution.

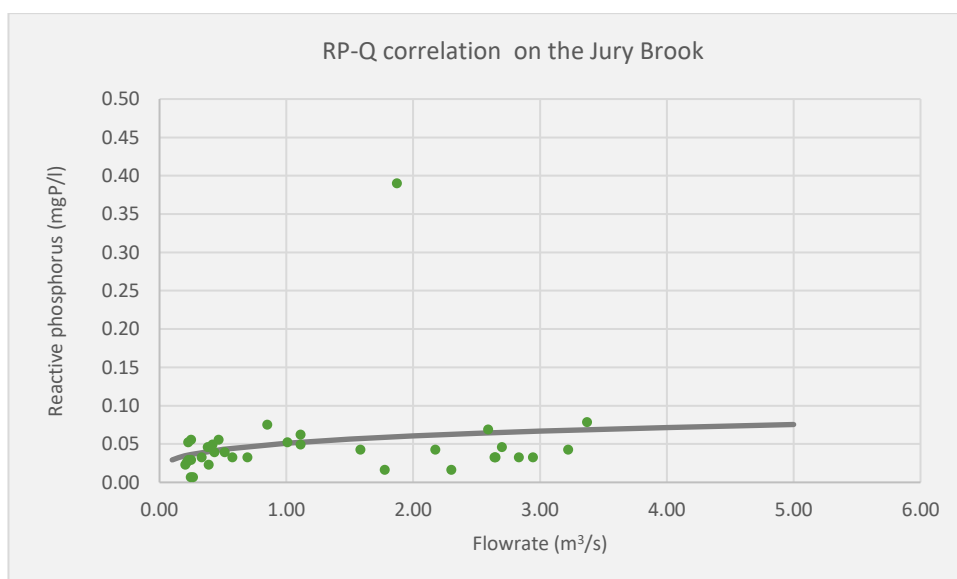


Figure 9: RP-flowrate correlation for the Jury Brook. The grey line represents the best-fit LAM model.

There is, however, a notable outlier recorded on 03-May. This result is explained by a pulse of water following heavy rain, which washed recently spread slurry from adjacent fields (all noted in Epicollect).

In order to better understand the overall picture for RP, figure 10 presents annual average RP values on the sub catchment map.

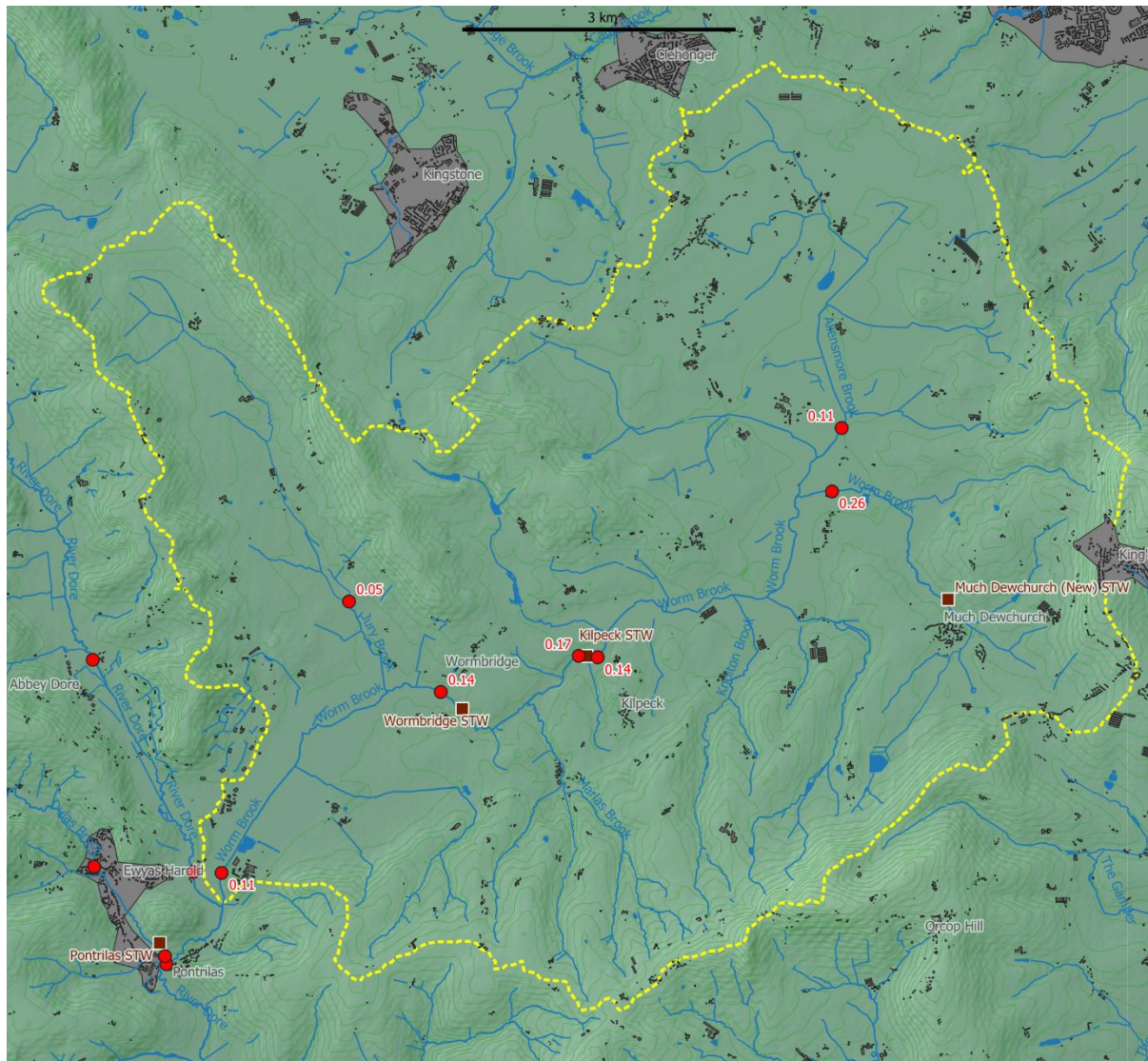


Figure 10: annual average RP (mgP/l) at sample locations across the Worm sub catchment.

The Allensmore Brook is diffuse dominated, with relatively little fixed-flow RP, and could be seen to represent the 'diffuse background' for the Worm sub catchment at 0.11mgP/l. The exception is the Jury Brook which is topographically more like the adjacent Dore valley than the Worm and has a significantly lower mean RP level (0.05mgP/l), similar to the Dore above Peterchurch. As discussed above, the highest RP levels are to be found immediately downstream of Much Dewchurch STW (0.26mgP/l). The fixed-flow signal progressively dilutes as it flows down to Kilpeck STW (0.14mgP/l), where it is topped up again (0.17mgP/l). We then find a progressive fall moving downstream to Sawmills Footbridge (0.11mgP/l). The fixed flow signal is still present at this point and equates to an RP load of around 1.8kgP/day. To which we must add in the region of 6kgP/day from diffuse sources.

Conclusion

- Flow modelling
 - The volumetric flow data from Marstow Mill on the Garren has proved to be a useful surrogate for flowrates within the Worm sub catchment as evidenced, for example, by the tight data spread on the LAM analysis of figure 8.
- Data outliers
 - Both EC and pH data show multiple temporally correlated spikes. On one occasion there is a correlation between EC and pH signals at the same location.
 - Turbidity showed a single spike at one location where a severe, localised sediment load was observed.
 - RP on the Jury Brook also showed a significant spike.
 - All of the above are clear indications of episodic pollution arising from sources throughout the catchment, much of which can only be attributed to sloppy farming practices.
- Turbidity
 - A fuller analysis of turbidity has been deferred until we have a complete year's data.
- Reactive phosphorus
 - The Jury Brook has significantly lower RP than the rest of the Worm sub catchment. LAM analysis shows only diffuse source RP, with no fixed flow signal. The annual mean concentration (0.05mgP/l) falls within the range of WFD *good* status.
 - The rest of the sub catchment (Worm and Allensmore) shows a mix of both diffuse and point source RP. The background, diffuse source RP is generally in the region of 0.10mgP/l in all locations, which corresponds to WFD *moderate* status. However, fixed flow RP, principally from STWs, take annual average values much higher than this at most locations.
 - Effluent from Much Dewchurch STW is massive RP load for the tiny upper Worm, taking the annual mean concentration up to 0.26mgP/l – corresponding to WFD *poor* status.
 - Effluent from Kilpeck STW causes an increase in RP concentration from 0.14mgP/l to 0.17mgP/l.
- STW effluent
 - The RP loads attributed to Much Dewchurch and Kilpeck STWs are but one small component of the pollution from these sources. Impact on this relatively small ecosystem is undoubtedly much wider than an RP concentration.
 - Significant volumes of untreated sewage are also flowing from Much Dewchurch.

References

Green, G. (2023). *Tech note: Dore 2023*.